

Research Interests

Self-Supervised Learning, World Models, Control and Planning, Efficient Architecture

Education

2023–present **Ph.D. in Data Science, New York University, USA.**

○ Research Focus: Self-Supervised Learning, World Models, Control and Planning

○ Advisor: [Yann LeCun](#)

○ Affiliation: [CDS](#), [CILVR](#)

2020–2023 : **B.A. in Mathematics with High Honors, Courant Institute, New York University, USA.**

○ Latin Honors: Magna Cum Laude with Minor in Computer Science

Publications

Conference Papers

2026 **Rectified LpJEPa: Joint-Embedding Predictive Architectures with Sparse and Maximum-Entropy Representations .**

Yilun Kuang, Yash Dagade, Tim G. J. Rudner, Randall Balestrierio, Yann LeCun.

International Conference on Machine Learning (ICML), 2026

2025 **Customizing the Inductive Biases of Softmax Attention using Structured Matrices.**

Yilun Kuang, Noah Amsel, Sanae Lotfi, Shikai Qiu, Andres Potapczynski, Andrew Gordon Wilson.

International Conference on Machine Learning (ICML), 2025

2024 **Bayesian Optimization of Antibodies Informed by a Generative Model of Evolving Sequences.**

Alan Nawzad Amin, Nate Gruver*, **Yilun Kuang*** (equal contribution), Yucen Lily Li*, Hunter Elliott, Aniruddh Raghu, Calvin McCarter, Peyton Greenside, Andrew Gordon Wilson.

International Conference on Learning Representations (ICLR), 2025, **Spotlight**

2024 **Unlocking Tokens as Data Points for Generalization Bounds on Larger Language Models.**

Sanae Lotfi*, **Yilun Kuang*** (equal contribution), Brandon Amos, Micah Goldblum, Marc Finzi, Andrew Gordon Wilson.

Neural Information Processing Systems (NeurIPS), 2024, **Spotlight**

2023 **Non-Vacuous Generalization Bounds for Large Language Models.**

Sanae Lotfi*, Marc Finzi*, **Yilun Kuang*** (equal contribution), Tim G. J. Rudner, Micah Goldblum, Andrew Gordon Wilson.

International Conference on Machine Learning (ICML), 2024

2023 **Learning Efficient Coding of Natural Images with Maximum Manifold Capacity Representations.**

Thomas Yerxa, **Yilun Kuang**, Eero Simoncelli, SueYeon Chung.

Neural Information Processing Systems (NeurIPS), 2023

Workshop Papers

2026 **LpWM: A Case for Sparse Representations in World Models.**

Yilun Kuang, Yash Dagade, Quentin Le Lidec, Lucas Maes, Randall Balestrierio, Yann LeCun.

Workshop on World Models at Chicago Booth

- 2025 **Radial-VCReg: More Informative Representation Learning through Radial Gaussianization.**
Yilun Kuang, Yash Dagade, Deep Chakraborty, Erik Learned-Miller, Randall Balestrieri, Tim G. J. Rudner, Yann LeCun.
 NeurIPS 2025 Workshop: Unifying Representations in Neural Models & Symmetry and Geometry in Neural Representations
- 2024 **Bayesian Optimization of Antibodies Informed by a Generative Model of Evolving Sequences.**
 Alan Nawzad Amin, Nate Gruver*, **Yilun Kuang*** (equal contribution), Yucen Lily Li*, Hunter Elliott, Aniruddh Raghu, Calvin McCarter, Peyton Greenside, Andrew Gordon Wilson.
 NeurIPS 2024 Workshop: AI for New Drug Modalities, **Spotlight & Outstanding Poster Award**
- 2024 **Unlocking Tokens as Data Points for Generalization Bounds on Larger Language Models.**
 Sanae Lotfi*, **Yilun Kuang*** (equal contribution), Brandon Amos, Micah Goldblum, Marc Finzi, Andrew Gordon Wilson.
 ICML 2024 Workshop: Theoretical Foundations of Foundation Models, **Best Paper Award**
- 2023 **Unsupervised Learning on Spontaneous Retinal Activity Leads to Efficient Neural Representation Geometry.**
 Andrew Ligeralde*, **Yilun Kuang*** (equal contribution), Thomas Edward Yerxa, Miah N Pitcher, Marla Feller, SueYeon Chung.
 NeurIPS 2023 Workshop: Unifying Representations in Neural Models (UniReps)
- 2023 **Non-Vacuous Generalization Bounds for Large Language Models.**
 Sanae Lotfi*, Marc Finzi*, **Yilun Kuang*** (equal contribution), Tim G. J. Rudner, Micah Goldblum, Andrew Gordon Wilson.
 NeurIPS 2023 Workshop: Self-Supervised Learning & Mathematics of Modern Machine Learning (M3L)
[Poster](#)
- 2023 **Learning a Visual Representation by Maximizing Manifold Capacity.**
 Thomas Yerxa, **Yilun Kuang**, Eero Simoncelli, SueYeon Chung.
 Computational and Systems Neuroscience (COSYNE), 2023

Work Experience

- May – Aug 2026 **Research Intern at AMI Labs.**
 ○ Mentor: [Yann LeCun](#)
 ○ Work on JEPA World Models.
- June – Sep 2022 **Research Intern at Flatiron Institute, Simons Foundation.**
 ○ Mentor: [SueYeon Chung](#)
 ○ Develop a state-of-the-art self-supervised learning algorithm for vision.
 ○ 1 publication in NeurIPS and 1 poster presentation in COSYNE.
- Jan 2023 **Open Source Contributor at MosaicML.**
 ○ Implement a distributed training pipeline for efficient deep neural network training. [Open source contribution](#) available in the Composer Library documentation.

Fellowships & Awards

- 2023 – 2028 **Center for Data Science Fellowship.**
 ○ Awarded the NYU Center for Data Science Fellowship for five years and the Data Science Supplementary Fellowship Grant
- May 2023 **Nicholas and Andrea Ferrara Research Scholar, Dean's List (2020-2022), DURF Grant Recipient, Best Presenter at NYU Undergraduate Research Conference.**
- May 2023 **Meritorious Winner in 2021 Mathematical Contest in Modeling (MCM) (Top 7%).**

Teaching Assistantship

- Fall 2025 **Section Leader**, DS-GA 1018: Probabilistic Time Series Analysis, New York University.

- Fall 2024 **Grader**, CSCI-GA 2565: Machine Learning, New York University.
- Spring 2024 **Grader**, DS-GA 1003: Machine Learning, New York University.
- Fall 2023 **Section Leader & Grader**, DS-GA 1011: Natural Language Processing with Representation Learning, New York University.
- Spring 2022 **Grader**, DS-GA 1012: Natural Language Understanding and Computational Semantics, New York University.

Summer School

- Summer 2020 **Neuromatch Academy in Computational Neuroscience**, Virtual.
- Summer 2020 **Summer Session in Mathematics**, Harvey Mudd College, CA.

Invited Talks

- Feb 2026 **Rectified LpJEPA: Sparse and Maximum-Entropy JEPA**, Shanghai AI Lab.

Position of Responsibility

- Conference **Reviewer**, NeurIPS 2024, NeurIPS 2025 Datasets and Benchmarks Track, ICML 2025-2026, ICLR 2025-2027, AISTATS 2026, AAAI 2026.
- Workshop **Reviewer**, NeurIPS 2023 Workshop: Self-Supervised Learning - Theory and Practice, ICML 2024 Workshop: Theoretical Foundations of Foundation Models (TF2M), NeurIPS 2024 Workshop: Adaptive Foundation Models: Evolving AI for Personalized and Efficient Learning (AFM), NeurIPS 2025 Workshop: Unifying Representations in Neural Models (UniReps), NeurIPS 2025 Workshop: Symmetry and Geometry in Neural Representations (NeurReps), ICLR 2026 Workshop: Scientific Methods for Understanding Deep Learning (Sci4DL), Chicago Booth 2026 Workshop on World Models, NeurIPS 2026 Workshop: Symmetry and Geometry in Neural Representations (NeurReps).

Media Coverage

- March 2026 **New Representation Learning Technique Matches the Efficiency of the Human Brain**, NYU Center for Data Science.
- Sep 2025 **How a Little Known Math Paper Inspired New Attention Mechanisms**, NYU Center for Data Science.
- June 2024 **Do Large Language Models Really Generalize? This Paper Says Yes**, NYU Center for Data Science.

Computer skills

- Programming Languages Python, C++, C, Java, Matlab, Julia, SQL, Bash, Vim, LaTeX
- Machine Learning PyTorch, Jax, Hugging Face, TensorFlow, Scikit-Learn, GPyTorch, NumPy, Pandas
- Cloud Services/Other AWS, GCP, Docker, Kubernetes, Vim, Git, MicroService, CUDA, OpenMP, MPI